

Data modeling: CSCI E-106

Evaluation Metrics for Confusion Matrix

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We use a confusion matrix to split the predictions into True Positives, False Positives, True Negatives, and False Negatives. In our case, True Positives are the applicants that were correctly identified as being "Good," False Positives are the applicants that were incorrectly identified as being "Good," True Negatives are the applicants that were correctly identified as "Bad," while False Negatives are the applicants that were incorrectly identified as "Bad."

Actual	Predicted	
	Positive	Negative
Positive	True Positive	False Negative
Negative	False Positive	True Negative

Evaluation Metrics for Confusion Matrix, *cont'd*

Accuracy: what percentage of the predictions were correct?

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives}}$$

Precision: of all the Good predictions, how many were actually Good?

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall: how many of the applicants with a Good credit rating were correctly identified?

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Evaluation Metrics for Confusion Matrix, *cont'd*

F1 Score: the harmonic mean of precision and recall, which is a common way to merge the two scores into one, making it easier to compare models.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$